Yujun Zhao

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School of Instrumentation & Optoelectronic Engineering, Beihang University, Beijing, China

EDUCATION

Beihang University *Sep 2024 – Present*

Master of Instrumentation Science and Technology

- **GPA:** 3.74/4.0
- **Core Courses:** Remote Sensing Image Interpretation (98), Measurement Uncertainty: Principles and Applications (98), Detection and Processing of Weak Signals (91), Theory and system analysis of electro-optical imaging and detection (90).

Changchun University of Science and Technology *Sep 2020 – Apr 2024*

Bachelor of Measurement and Control Technology

- **GPA:** 3.92/5.0
- **Core Courses:** Advanced Mathematics (99), Complex Function and Integral Transforms (98), Physical Optics (97), Digital Image Processing (96), Linear Algebra (95), Signals and systems (95).
- **Awards:** National Scholarship, Jilin Provincial Government Scholarship, Presidential Scholarship.

RESEARCH INTERESTS

Limb-Imaging Spectroscopy
Hyperspectral Remote Sensing
Atmospheric Radiative Transfer Modelling
Spectroscopic Instrument Calibration and Validation
Remote Sensing of Atmospheric Composition

PUBLICATION & CONFERENCE PAPERS

[1] **Zhao, Y.**, Jia, G., Li, Z., Ma, X., Xu, J., and Zhao, H.: Simulation and Validation of UV–Visible Limb-Scattered Imaging Spectroscopy Based on SASKTRAN-HR, EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026, EGU26-3334, https://doi.org/10.5194/egusphere-egu26-3334, 2026.

[2] Ma, X., Zhao, H., Jia, G., Xu, J., and **Zhao, Y.**: DeepOE: A physics-informed neural network for atmospheric temperature profile retrieval, Atmospheric and Oceanic Science Letters, 100784, doi:10.1016/j.aosl.2026.100784, 2026.

RESEARCH EXPERIENCES

Simulation and Validation of Limb-Scattering Imaging Spectroscopy

Project Leader *Sep 2025 - Present*

- Developed a radiative transfer simulation framework for UV–visible limb-scattering hyperspectral observations using the SASKTRAN-HR engine.
- Specified the atmospheric state based on CAMS reanalysis data and the MSIS-90 atmospheric model.
- Full-scene simulation including space-view, limb-view, and surface-view radiance components
- Performed validation of the atmospheric simulation against OSIRIS Level 1 single-scan observations.
- Performed validation of the surface simulation against CHRIS Level 1 observations.

Ground-Based Self-Calibrating Infrared Spectrometer for Atmospheric Observation

Project Contributor *Oct 2024 - Dec 2025*

- Designed the mechanical structure of a radiometric calibration unit integrated with an MR304 Fourier Transform Infrared spectroradiometer.
- Assembled and commissioned a self-calibrating infrared spectrometer.
- Conducted blackbody-based two-point calibration experiments for the instrument.
- Processed the calibration data and achieved radiometric calibration accuracy better than 1.2 K at 300 K across the mid- and long-wave infrared bands.

Simulation of a Spatial Heterodyne Spectrometer (SHS) for Atmospheric Sounding

Undergraduate Thesis *Sep 2023 – Jun 2024*

- Developed a MATLAB-based simulation framework for an SHS, covering the forward process from input spectral radiance to interferogram generation.
- Implemented detector-noise modelling and interferogram reconstruction procedures, including phase correction and inverse Fourier transform.

SKILLS

Programming: Python · MATLAB
Scientific Tools: SASKTRAN · MODTRAN · LBLRTM · Fourier Transform Spectroscopy Processing · Linux System
Spectroscopic Instruments: MR304 Fourier Transform Infrared Spectroradiometer · Atmospheric Emitted Radiance Interferometer (AERI) · ASD Handheld Spectroradiometer (Field Spectral Measurements)
Additional Software: Microsoft Office (Word, PowerPoint, Excel) · Origin · ENVI
Certificates: IELTS (6.5)
Interests: Calligraphy · Fitness